

Heavy, Light, Hybrid

Running all three models of DNA replication side by side, and finding the one generation that actually decides between them.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/meselson-stahl/

1 Before you open anything

Answer from what you already know. Getting it right is not the point — having something on paper to argue with later is.

PREDICTION

E. coli is grown for many generations in ^{15}N until every DNA strand is heavy, then transferred to ^{14}N medium. Three models are on the table: conservative, semiconservative, dispersive. Counting the transfer as generation 0, **which is the first generation whose density-gradient tube can rule out the dispersive model?** Write a single generation number.

In one or two sentences, say what the tube would have to look like at that generation for dispersive to be eliminated.

2 Reading all three predictions at once

Open the demonstration. The row of buttons labelled **Generations in ^{14}N medium** sets the generation from 0 to 4. The **Predicted bands** table always lists all three models, whichever one is highlighted — that table is where you take your numbers from.

1. Leave the generation on **0**. Read the **Predicted bands** table and fill in the three generation-0 rows below. Use the **Density** and **Share** columns exactly as printed.
2. Click **1** (or press **Next generation**) and fill in the three generation-1 rows.
3. For each row, the number of bands is just how many lines that model occupies in the table.

Gen	Model	Bands	Density of each band (g/cm ³)	Share of each band
0	Conservative			
0	Semiconservative			
0	Dispersive			
1	Conservative			
1	Semiconservative			
1	Dispersive			

a. All three generation-0 rows are identical. Explain, in terms of what the culture had been doing before the transfer, why that had to be true – and say what it means for generation 0 as a test of the three models.

b. At generation 1, one model is the only one still placing DNA at 1.7240 g/cm³. Name it, and give the share the table predicts for that band.

c. At generation 1, two of the rows are identical to each other. Which two, and what does that identity prevent you from concluding?

Band darkness in the tubes is drawn proportional to the share of total DNA, so a small band is genuinely faint on screen. Read the printed percentage rather than judging by eye.

3 What changes and what refuses to move

Now step the generation across its whole range and watch two quantities at once: where a band sits, and how much DNA is in it. The semiconservative and dispersive rows behave in opposite ways, and that contrast is the entire experiment.

1. Set the generation to **1**, then **2**, **3**, **4** in turn.
2. At each generation read **Molecules** from the **Readouts** panel.
3. From the **Predicted bands** table, record the semiconservative *hybrid* band and the single dispersive band.

Gen	Molecules	Semiconservative hybrid band: density	Semiconservative hybrid band: share	Dispersive band: density	Dispersive band: share
1					
2					
3					
4					

a. Of the four band columns, two hold the same value at every generation. Name both, and for each say what the model requires that forces it to stay fixed.

b. The semiconservative hybrid band stays at one density but its share halves each generation. Write that share as a formula in the generation number n , and explain it by counting molecules: how many hybrid molecules exist at generation n , out of how many total?

c. The dispersive band keeps 100% of the DNA but its density falls. Write the fraction of a dispersive molecule's nitrogen that is ^{15}N as a formula in n , and check it against the ^{15}N share readout.

d. State in one sentence the rule that separates the two models: under one of them the band count changes, under the other the band *position* changes.

e. Return to your prediction in part 1. Which generation actually settles it? If your number was different, say what you had assumed about the dispersive prediction that the table does not support.

4 Reading a tube backwards

An exam question rarely hands you the model and the generation. It hands you a tube. Work each of these out on paper first, then set the demonstration to check.

What the tube shows	Model	Generation	How you decided
One band, 100% of the DNA, at 1.7135 g/cm ³			
Two bands: 12.5% at 1.7170 and 87.5% at 1.7100			
Two bands: 25% at 1.7240 and 75% at 1.7100			

a. A shortcut that sounds reasonable: *one band means dispersive, two bands means semiconservative*. Set the generation to **1** with **Semiconservative** selected under **Molecules drawn above**, and read the **Bands** readout. What does it say, and where does the shortcut break?

b. Tick **Reveal what Meselson & Stahl actually saw** and step from generation 1 to generation 2. Write down what was observed at each, then explain why the generation-1 observation, on its own, cannot establish semiconservative replication.

c. Read the **Where the model simplifies** panel, then go to generation 4 and look at the dispersive tube. Give one reason a real experiment run far past generation 2 would have been a weaker argument than generation 2 itself.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself – and where you were off, write down which step went wrong. Take the reference densities as 1.7100 g/cm^3 for all- ^{14}N DNA and 1.7240 g/cm^3 for all- ^{15}N DNA, with density straight-line in the fraction of ^{15}N .

1. Replication is semiconservative. A fully ^{15}N culture has grown three generations in ^{14}N . State the total number of molecules, the number of hybrid molecules, and the density and share of every band in the tube.

2. Suppose instead that replication were dispersive. After four generations in ^{14}N , what fraction of each molecule's nitrogen is ^{15}N ? Use the linear relationship to calculate the band density to four decimal places.

3. Suppose instead that replication were conservative. After four generations, give the share of the DNA sitting at 1.7240 g/cm^3 as an exact fraction and as a percentage. The demonstration rounds shares to one decimal place – state what it will display and why that differs from your exact value.

4. At every generation, and under all three models, the original ^{15}N accounts for the same fraction of the total DNA. Explain why that must be so, and then explain how a caesium chloride gradient can still tell the three models apart despite it. No calculation is needed.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Hold the generation at 3 and click through all three buttons under **Molecules drawn above**. Each model draws the same eight molecules differently. Describe, strand by strand, what makes the dispersive picture impossible to sort into “old” and “new” strands.
- Under dispersive replication the drawn strands are built from 16 blocks. Track how many blocks stay heavy at each generation and work out what physically has to happen to a strand for that pattern to be produced.
- Step through generations 3 and 4 with the observations revealed. Meselson and Stahl followed the culture out that far even though generation 2 had already decided the question. Argue for what those extra generations were worth.